

Marshall Excelsior

Gas Connections

MEGR-6120 GAS REGULATOR

Instruction Manual- Look Inside For:

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Marshall Excelsior Company

Marshall, MI 49068
269-789-6700
FAX 269-781-8340

www.marshallexcelsior.com

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ATTENTION:

The installation of this device must be in accordance with NFPA standards No. 54 & 58, along with all federal, state, and local codes. All personnel installing this device should be trained on the hazards associated with combustible gases and fluids. MEC assumes no responsibility from damage or injury that may result from servicing or the improper installation of this regulator.

OPERATION:

The MEGR-6120 is a high capacity, pounds to pounds, industrial gas regulator. It is designed to conform to UL Standard 144 for use with LP Gas. The maximum supply pressure is 250 PSIG. The maximum output pressure is printed on the regulator nameplate. This regulator is not intended for use in pressure applications below 3 PSIG. The operating temperature range is -40° to +200° F.

INSTALLATION:

Ensure all gas lines are free from dirt and scale. Shut off the gas. The MEGR-6120 can be mounted at any angle without affecting operation. Install the regulator as close as possible to the instrument that it is to supply. Avoid using undersized fittings that will limit the flow through the regulator and cause downstream pressure drop. Apply a minimum amount of pipe compound to the male threads of the inlet and outlet piping. The words "IN" and "OUT" are cast into the body of the regulator to indicate the direction of flow. The regulator must be oriented in the correct flow direction or internal damage to the regulator may occur. The two 1/4" NPT gauge ports marked "GA" can also be used as optional output ports, but with decreased flow delivery. Before putting the regulator into service for the first time, relieve compression on the range spring by turning the "T" handle counter clockwise. After gas is applied to the regulator, check installation to ensure that there is a leak tight seal. To operate, turn the "T" handle slowly in a clockwise direction until the required downstream pressure is obtained.

CAUTION:

The MEGR-6120 is a non-relieving regulator and thus cannot exhaust excess pressure at the regulator. In order to establish an accurate output pressure it may be necessary to cycle the downstream flow or to set the regulator at a slight flow.

MAINTENANCE:

Because the MEGR-6120 is assembled and tested at the factory to conform to UL Standard 144, any maintenance or repair should be in accordance with this listing as well as any other applicable regulations. Regulator parts showing wear should be replaced as necessary. The procedure below describes how to disassemble the MEGR-6120 for parts replacement and inspection. Assembly is the reverse of this procedure.

1. Turn the adjustment screw counterclockwise to remove spring tension.
2. Loosen the six build screws and separate the bonnet from the body.
3. Remove the spring guide and spring.
4. Unscrew the diaphragm assembly from the pintle assembly.
5. Loosen, by turning counterclockwise, and remove the O-Ring and bottom plug.
6. Remove the pintle spring and pintle assembly.

HIGH PRESSURE REGULATORS

MEGR-6120 & MEGR-6121 SERIES



MEGR-6120 Series



MEGR-6121 Series



Adjustable Configurations	
Part No.	Description
MEGR-6120-30	High Pressure Adjustable Reg 1-30 PSI - 1/4" FNPT Inlet/Outlet
MEGR-6120-60	High Pressure Adjustable Reg 1-60 PSI - 1/4" FNPT Inlet/Outlet
MEGR-6120-100	High Pressure Adjustable Reg 1-100 PSI - 1/4" FNPT Inlet/Outlet

Non-Adjustable Configurations	
Part No.	Description
MEGR-6121-05	High Pressure Fixed Reg 5PSI - 1/4" FNPT Inlet/Outlet
MEGR-6121-10	High Pressure Fixed Reg 10PSI - 1/4" FNPT Inlet/Outlet
MEGR-6121-20	High Pressure Fixed Reg 20PSI - 1/4" FNPT Inlet/Outlet
MEGR-6121-30	High Pressure Fixed Reg 30PSI - 1/4" FNPT Inlet/Outlet
MEGR-6121-40	High Pressure Fixed Reg 40PSI - 1/4" FNPT Inlet/Outlet
MEGR-6121-50	High Pressure Fixed Reg 50PSI - 1/4" FNPT Inlet/Outlet
MEGR-6121-60	High Pressure Fixed Reg 60PSI - 1/4" FNPT Inlet/Outlet

Application:

The MEGR-6120 Series are UL LISTED high-pressure regulators that meet a variety of applications for liquid or vapor service. The compact body design makes these regulators particularly useful in installations with space limitations. The basic MEGR-6120 Series regulators come equipped with a handwheel adjustment. The non-adjustable ME6121 Series provides a tamper-resistant spring case and one of seven fixed setpoints: 5, 10, 20, 30, 40, 50, or 60 PSI.

The MEGR-6120 & MEGR-6121 Series are UL listed as high pressure, non-relief regulators. Both MEGR-6120 & MEGR-6121 Series regulators contain brass materials that are **not** compatible with anhydrous ammonia service.

MEGR-6120 & MEGR-6121 Specifications:

Type:	Adjustable or Fixed / PSI
Max. Inlet Pressure:	250 PSIG (17,2 bar)
Inlet Connection:	1/4" FNPT
Outlet Connections:	1/4" FNPT
Gauge Port:	1/4" FNPT
Exterior Finish:	Red Wet Coat
Diaphragm:	Fabric Reinforced Buna N
Body/ Bonnet Material:	Die Cast Aluminum
Liquid Capacity:	3-5 GPH
Listings:	UL LISTED / UL 144

CAPACITIES: BTU/H LPG (VAPOR)

Outlet Set Point	Inlet Pressure	BTU / HR
15 PSI	50 PSI	850,000
15 PSI	100 PSI	1,700,000
15 PSI	150 PSI	2,500,000
20 PSI	50 PSI	900,000
20 PSI	100 PSI	1,800,000
40 PSI	100 PSI	1,500,000
40 PSI	150 PSI	2,000,000
50 PSI	100 PSI	1,300,000
50 PSI	150 PSI	1,800,000
50 PSI	200 PSI	2,300,000

*Approximate BTU/H vapor capacities taking 10-20% droop into consideration

Note: Side Outlet Connection Style (Plugged): 1/4" MNPT, pressure gauge (MEJ500 Series) can be installed